

FAITH OGUNDIMU

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[LinkedIn](#) | [GitHub](#) | [Website](#)

EDUCATION

Royal College of Surgeons Ireland (RCSI)

2025 - 2029

Doctor of Philosophy (PhD) in Cancer Genomics and Epigenomics, Machine Learning and Bioinformatics

- Investigating non-coding driver mutations in breast cancer using large-scale datasets (TCGA, ICGC, ENCODE) and ML-based integration of chromatin accessibility, mutation, expression and spatial transcriptomic data.
- Developing machine learning approaches including Hidden Markov Models and deep learning methods (convolutional neural networks, multilayer perceptrons) to prioritize candidate non-coding drivers from >5,000 breast cancer whole genomes; implementing CRISPR-based validation in breast cancer cell lines.
- Creating 3D breast tissue-specific regulatory maps by integrating single-cell ATAC-seq data with ENCODE regulatory elements to focus analysis on the 98% non-coding genome where driver mutations remain largely uncharacterized.

Dublin City University (DCU)

2021 - 2025

BSc. in Genetics and Cell Biology (GCB) | First Class Honours (1.1) | Ranked 2nd & Dean's Honour List

Final Dissertation: *Using neural networks and foundation models to understand gene regulation in the MCF-7 breast cancer cell line*

PROFESSIONAL EXPERIENCE

Beaumont RCSI Cancer Centre, Irish Cancer Society, Lung Health Check

January 2026 - March 2026

Bioinformatics Research Assistant – BRAND Study (NSCLC Liquid Biopsy & cfDNA Methylation Analysis)

- Contributed to Ireland's largest lung cancer pilot (€4.9M) by developing and iterating longitudinal cfDNA methylation workflows to quantify circulating tumour signal and link it to clinical outcomes, including tumour methylation fraction (TMF) estimation via semi-reference-free deconvolution (SRFD)
- Observed a significant post-immunotherapy TMF reduction (Wilcoxon $p=0.015$, effect size $r=0.85$) and found Δ TMF to be a near-significant independent predictor of survival in multivariable Cox regression ($HR=1.14$, $p=0.068$) adjusting for treatment, PD-L1, age, and ECOG.
- Built an end-to-end cfDNA methylation analysis and DMR discovery framework across 88 bisulfite-sequenced plasma samples (EBC, ~8.7M CpGs), combining count-aware region calling (DSS), interaction-based modelling for responder vs non-responder contrasts, and metilene-based Galaxy runs aligned to the TWIST pipeline; reduced marker space from >89,000 candidates to 100 high-information CpGs (mean $|\Delta\beta|=0.57$) and produced publication-style visual summaries and annotations for downstream interpretation.

University College Dublin

June 2025 - August 2025

Breakthrough Summer Research Scholar (BioSLATE - Biomarker Selection and Synthetic Lethality Analysis for Therapeutic

Exploration in high-grade serous ovarian cancer (HGSOC)) - [BioSLATE Website](#) | [GitHub \(BioSLATE\)](#) | [GitHub \(OncoSynth\)](#)

- Identified synthetic lethal targets in HGSOC, as measured by statistically significant CNA-CRISPR correlations across >500 patient tumours, by integrating TCGA CNA profiles, DepMap CRISPR screens and gene essentiality models within a high-throughput biomarker discovery pipeline; deployed results via an interactive Streamlit app for biomarker nomination and visual exploration.
- Built and scaled OncoSynth, a command-line multi-agent system using CrewAI to automate literature mining, druggability scoring, and clinical trial linkage, by developing modular predictive workflows for translational cancer target discovery.

Royal College of Surgeons Ireland (Final Year Dissertation)

January 2024 - March 2024

Visiting Student Researcher - [GitHub](#)

- Built a feed-forward neural network on the HPC MeluXina supercomputer to predict chromatin accessibility in MCF-7 cells from nine epigenetic features (>2.8M bins), achieving 98% accuracy and outperforming classical baselines and published models.
- Applied LOCO and SHAP analyses to rank feature importance and benchmarked against the GET foundation model, identifying H3K4me1 as the strongest enhancer-linked predictor independent of DNA sequence features.

Virtual Metabolic Human (Virtuome)

May 2024 - July 2024

Research Intern

- Performed biocomputational modelling and visualisation of >1000 drug-metabolism pathways, improving pipeline efficiency by ~45%, and co-authored a Coeliac Society report and infographic translating digital health data into patient insights.

Applied Process Company (APC) Ltd.

January 2024 - August 2024

Analytical Intern

- Independently executed plasmid DNA extraction and quality-control workflows for high-priority biopharma projects, completing ~80% of assays autonomously and ensuring GLP-compliant documentation and data integrity for downstream computational analysis with HPLCs and Agilent OpenLab ChemStation.

Royal Victoria Eye And Ear Hospital

July 2023 - September 2023

Histopathology Summer Intern

- Streamlined histopathology workflows and digital data tracking in Clinisys WinPath, reducing turnaround times by 60%, and conducted microtomy, H&E staining, and exome-sequencing analyses to support research on uveal melanoma prognostics in young patients.

HACKATHONS & RESEARCH/TECHNICAL PROJECTS

Google Gemini Live Agent Challenge (Live Agents Track) [GitHub](#) | [Live Link](#) | [Devpost](#)

April 2026

Relay – Real-Time Voice & Vision AI Lab Tutor for Electronics – Honorable Mention (out of ~12,000 participants)

- Built Relay, a real-time voice + vision AI lab partner that watches a user's breadboard via webcam to verify components, guide wiring, generate and debug Arduino code, and produce a 3D digital twin of the build, using a multi-model architecture on the Gemini Live API (gemini-2.5-flash-native-audio for live voice, gemini-2.5-flash for component ID, gemini-2.5-pro for code gen and Gemini Vision drawing assessment) deployed across two FastAPI microservices on Google Cloud Run with Vertex AI, Google ADK bidi WebSocket streaming, Firestore persistence, and a fully procedural Three.js sandbox with circuit validation.

Harvard Health Systems Innovation Lab (HSIL) Hackathon [GitHub](#) | [Live Link](#)

April 2026

CareSignal – Ambient AI Care Scribe for UK Domiciliary Care

- Built CareSignal, an ambient AI care scribe that converts carer-patient conversations into CQC-compliant notes (10 structured fields across 5 domains) in <60 seconds per visit (vs. 30–48 minutes manually) using Google Cloud Speech-to-Text, Gemini 2.5 Flash, and ElevenLabs TTS, and shipped it as a full-stack platform with Twilio SMS + GPS lone-worker alerts, a manager analytics dashboard with anomaly detection, CQC-ready PDF exports, and Birdie/PASS interoperability via FHIR R4 + SNOMED CT.

Hack-Nation 4th Edition x MIT Sloan AI Club (ElevenLabs Track) [GitHub](#) | [Live Link](#) | [Demo Video](#)

February 2026

HealthCallPilot - Agentic Voice-AI Appointment Orchestration System - Finalist (Top 5 of 1,500 participants)

- Developed HealthCallPilot, an agentic voice-AI system that autonomously books, cancels, and reschedules healthcare appointments via ElevenLabs Conversational AI + Twilio outbound calling + FastAPI tool-calling + Google Calendar APIs, with a parallel "swarm" provider outreach workflow, real-time WebSocket dashboards, waitlist/callback intelligence, and slot-ranking by time/distance/preference.

Google DeepMind Vibe Code with Gemini 3 Pro in AI Studio Hackathon

December 2025

RegulonForge - Multimodal Regulatory Genomics Simulator - Winner (50 winners selected out of 4,097 submissions)

[Writeup](#) | [Live Link](#) | [Demo Video](#)

- Built RegulonForge, an interactive multimodal regulatory genomics simulator on Gemini 3 Pro that pairs sequence analysis with genome browser image interpretation (UCSC, IGV) for vision-language reasoning, and shipped perturbation sandbox, sketch-to-sequence, and evolution-viewer modules for side-by-side wild-type vs. mutant comparison of motifs, TF binding logic, and regulatory network topology, plus a teaching mode with interactive quizzes.

Google

October 2025

2025 Google Ireland Hackathon by BGN - 2nd Place - [TestBuddy \(Demo Videos\)](#)

- Selected from 500+ applicants and co-developed TestBuddy, an AI-powered driving education platform achieving 2nd place, as measured by judges' scores on innovation, presentation and technical implementation, by integrating adaptive testing algorithms with Google Cloud Platform APIs (Maps, Vision AI, Cloud Run) and implementing real-time test booking functionality.

SKILLS

Bioinformatics & Data Science: NGS Analysis, Genomic Databases (TCGA, DepMap, ENCODE), Statistical Modelling, Bisulfite Sequencing Analysis, cfDNA/Liquid Biopsy, Methylation Analysis, Survival Analysis (Kaplan-Meier, Cox Regression) GWAS, Workflow Automation, Biostatistics, Data Visualisation (Matplotlib, Seaborn, ggplot2), Docker

Programming & Scripting: Python, R, Bash, Perl, Markdown, Typescript

Machine Learning & AI Frameworks: Scikit-learn, TensorFlow/Keras, XGBoost, PyTorch, LangChain, Google Cloud Platform (GCP), Vertex AI, Multi-Agent Architectures (CrewAI, Google ADK), NMF (Non-negative Matrix Factorization), NNLS Deconvolution, Retrieval-Augmented Generation (RAG)

Notebook & App Development: Streamlit, Jupyter, Colab, FastAPI, React.js, Material UI, Firebase, Gemini APIs

Experimental and Lab Techniques: MiniPrep, NanoDrop 8, Biological Safety Cabinet, SDS-PAGE, HPLC (AEX), SoloVPE, ELISA, pDNA and gDNA Extraction, qPCR

Languages: English (Native), French (Professional Working Proficiency), Irish (Conversational)

GRANTS & SCHOLARSHIPS

[France Excellence Research Residency Grant](#) (€1,704, **French Embassy in Ireland**) – Competitively awarded to 9 of all applicants nationally; funding a 1-month research residency at Institut Curie (CUBIC, Prof. Nicolas Servant) for CNV-aware chromatin topology analysis and AI-ready regulatory map construction in breast cancer.

[Government of Ireland \(GOI\) Postgraduate Scholarship](#) (€136,000, **18% success rate**) – Nationally competitive award recognising exceptional research potential.

RCSI Top-Up Research Grant (€10,000) – Selective supplement to GOI Scholarship supporting CRISPR-based validation of computationally predicted non-coding driver mutations in breast cancer.

[Breakthrough Cancer Research Summer Scholarship](#) (€3,000) – National cancer research award supporting hands-on projects in poor-prognosis cancers.

CONFERENCES, PRESENTATIONS & TRAINING

Imperial College London Robotics Summer School (TBD July 2026): Foundational robotics, robot learning, and hands-on group project at the Hamlyn Centre for Robotic Surgery.

Panel Speaker, UCD Women+ In STEM Society Conference (sponsored by AMD) (Apr 2026): spoke on entering STEM, academia vs. industry, and the future of computational biology

RCSI Research Day - Poster (Mar 2026): Using neural networks and foundation models to understand gene regulation in the MCF-7 breast cancer cell line

NVIDIA Deep Learning Institute (DLI) Workshop (Nov 2025): Introduction to Deep Learning, Autoencoders and PyTorch — CECAM-IRL, University College Dublin

BRS Research Day, DCU (Apr 2025) – Poster: Using neural networks and foundation models to understand gene regulation in the MCF-7 breast cancer cell line

Conway Institute Research Festival, UCD (Oct 2025) – Poster: BioSLATE – Biomarker Selection and Synthetic Lethality Analysis in HGSOc.

PUBLICATIONS

Ogundimu F. & Furney SJ: Deciphering the Non-coding Cancer Genome through Deep Learning and Multiomic Integration (*in revisions - npj Genomic Medicine*)

O'Reilly D, et al.: Biomarkers of Response to Immunotherapy in Advanced Non-small cell lung cancer using Dynamic breath and blood analysis (BRAND) (*in preparation - targeted journal; Journal of Thoracic Oncology*)

Thinnes C. et al.: DrugMap: A genome-scale visualisation of human drug metabolism (*in preparation*)

ACADEMIC DISTINCTIONS & AWARDS

Conway Institute Research Festival – Poster Commendation (Perfect 30/30)

Breakthrough Cancer Research – Best Lay Presentation Award (2025)

BSc. Genetics & Cell Biology – First-Ranked (Years 1–2), Second-Ranked (Years 3–4), Salutatorian & Dean's Honours List

Bsc. Faculty of Science & Health - Undergraduate Academic Achievement Award

FSH Academic Scholarship